

Determining Coastal Flood Risk with Data and Graphical Analysis of Precipitation and Tide Data

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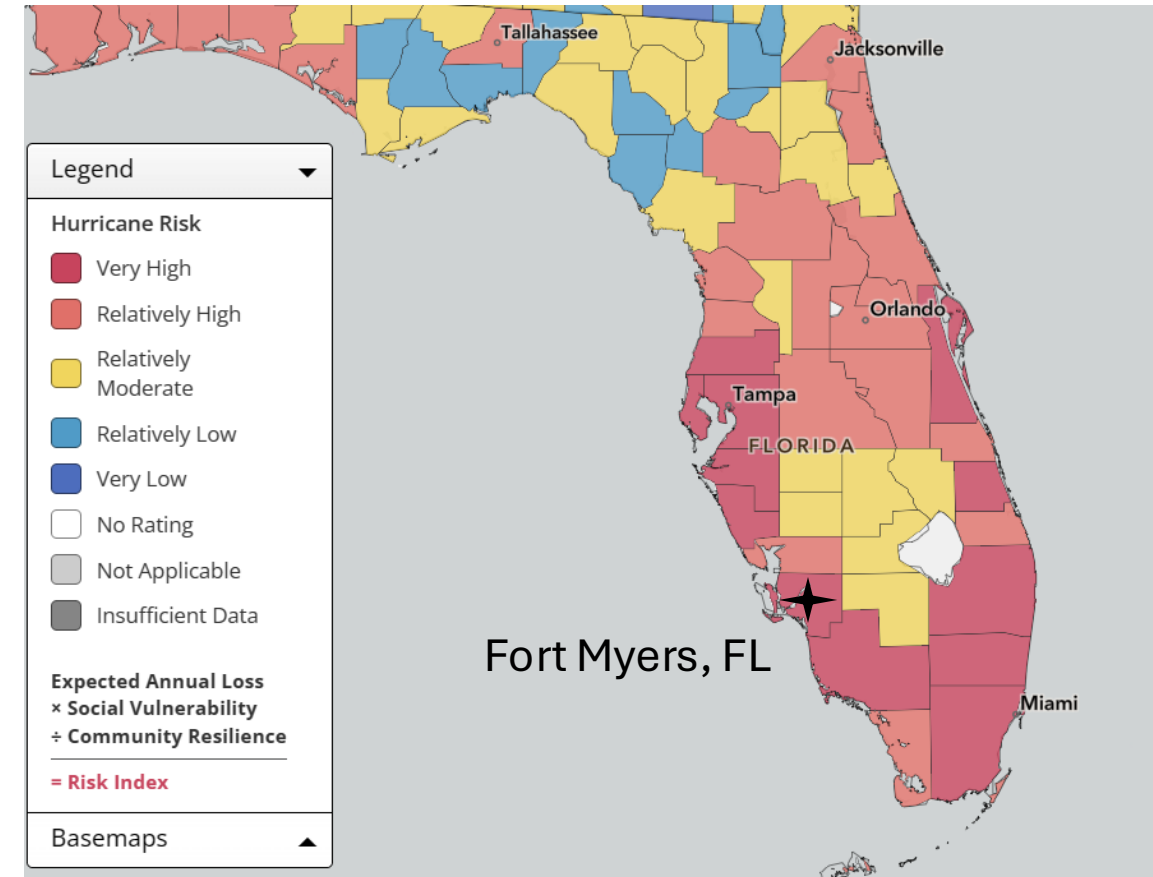
Introduction

- Increases in the frequency and intensity of extreme weather events can lead to property loss, cause costly disruptions to society, and reduce the affordability of insurance (Ebi et. al., 2018).
- High tide flooding regionally occurs more often in certain seasons and during certain years, which is important for awareness and preparedness purposes (Sweet et al. 2018).
- Hypothesis: There is a higher chance for flooding during high tide and extreme precipitation events, and these events would happen during months of heavier precipitation and during high tide.

[Pictures of Hurricane Ian Aftermath in Fort Myers Show Destruction of Area \(newsweek.com\)](https://www.newsweek.com/pictures-hurricane-ian-aftermath-fort-myers-show-destruction-area-1500000)

Reasons for Choosing Fort Myers, FL

- Location
 - Fort Myers, FL is on the coast of Florida.
 - Relatively flat terrain (elevation is near sea level)
 - Temperatures are usually too warm for snow, which would make working with our dataset easier, since we do not need to account for snowmelt
- Availability and Completeness of Dataset



Southwest Florida is known to be very vulnerable to flooding due to tropical storms and hurricanes.



Source Data Overview

- Location: Fort Myers, FL
- Field observations collected from the National Oceanic and Atmospheric Administration (NOAA) for the past 20 years
 - Daily Precipitation Data (in.)
 - 25 Weather Stations
 - Date range: January 1, 2003 – December 31, 2023
 - A single .CSV file with headers including “Station”, “Station Name”, “Elevation”, “Latitude”, “Longitude”, “Date”, and the “PRCP”, and “PRCP_Attributes”
 - Hourly Tide Data (ft. above/below MSL)
 - Date range: January 1, 1998 – December 31, 2023
 - Multiple Yearly .CSV files with headers including “Date”, “Time”, “Predicted (ft)”, “Preliminary (ft)” (usually empty data column), and “Verified (ft)”
 - Note: Yearly files were combined into one file to conduct data and graphical analysis

Precipitation Station Locations at Fort Myers, FL

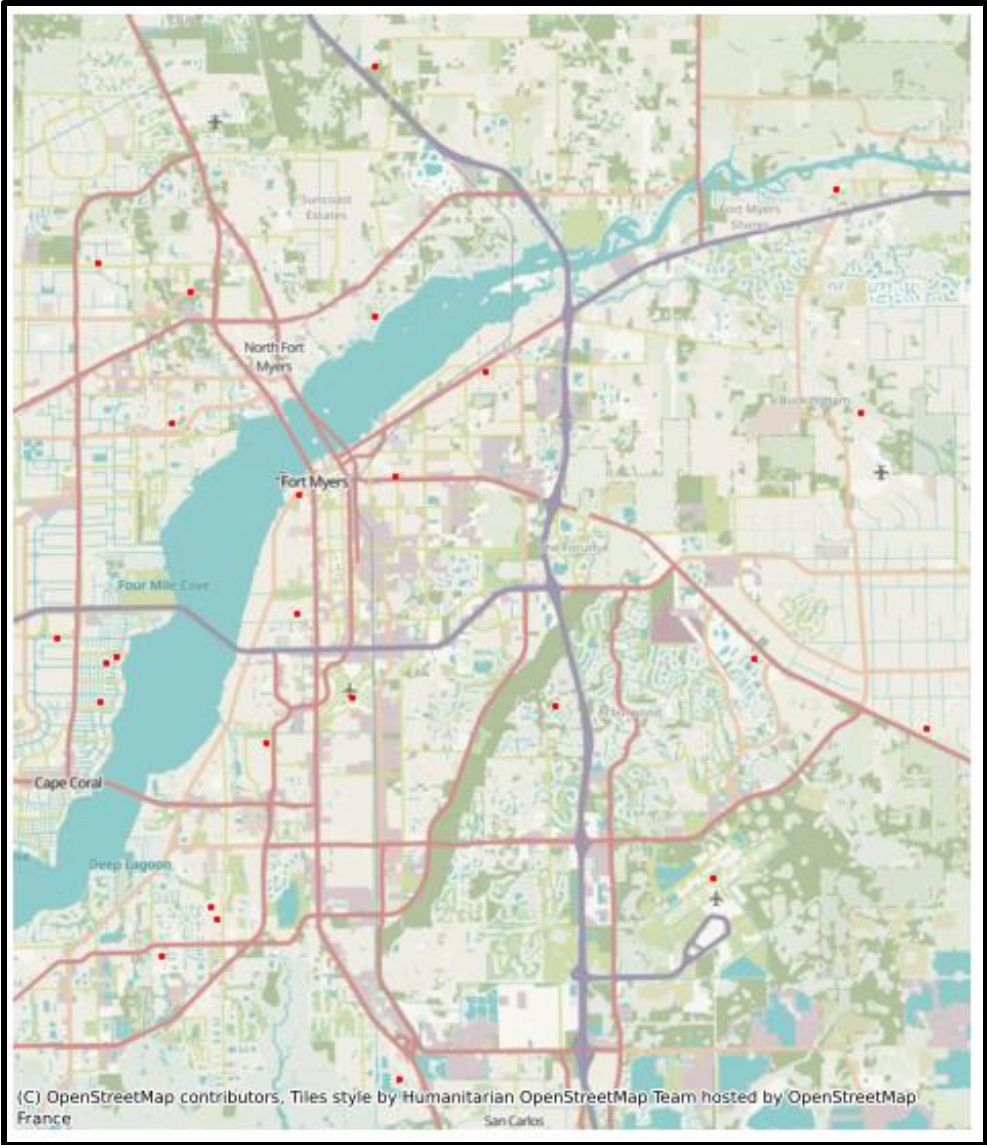
STATION	Elevation	STATION	Elevation
US1FLLE0001	3	US1FLLE0049	2.7
US1FLLE0003	7	US1FLLE0052	3
US1FLLE0004	0.3	US1FLLE0055	4
US1FLLE0005	11	US1FLLE0056	7.6
US1FLLE0006	13.1	US1FLLE0058	2.1
US1FLLE0011	17.1	US1FLLE0062	0.9
US1FLLE0017	1.2	US1FLLE0065	4.9
US1FLLE0023	4	US1FLLE0072	2.7
US1FLLE0029	1.2	US1FLLE0074	0.3
US1FLLE0037	3.7	US1FLLE0076	2.4
US1FLLE0039	6.1	USW00012835	3.8
US1FLLE0046	2.1	USW00012894	8.2
US1FLLE0047	3		

Relatively flat terrain (elevation is near sea level)

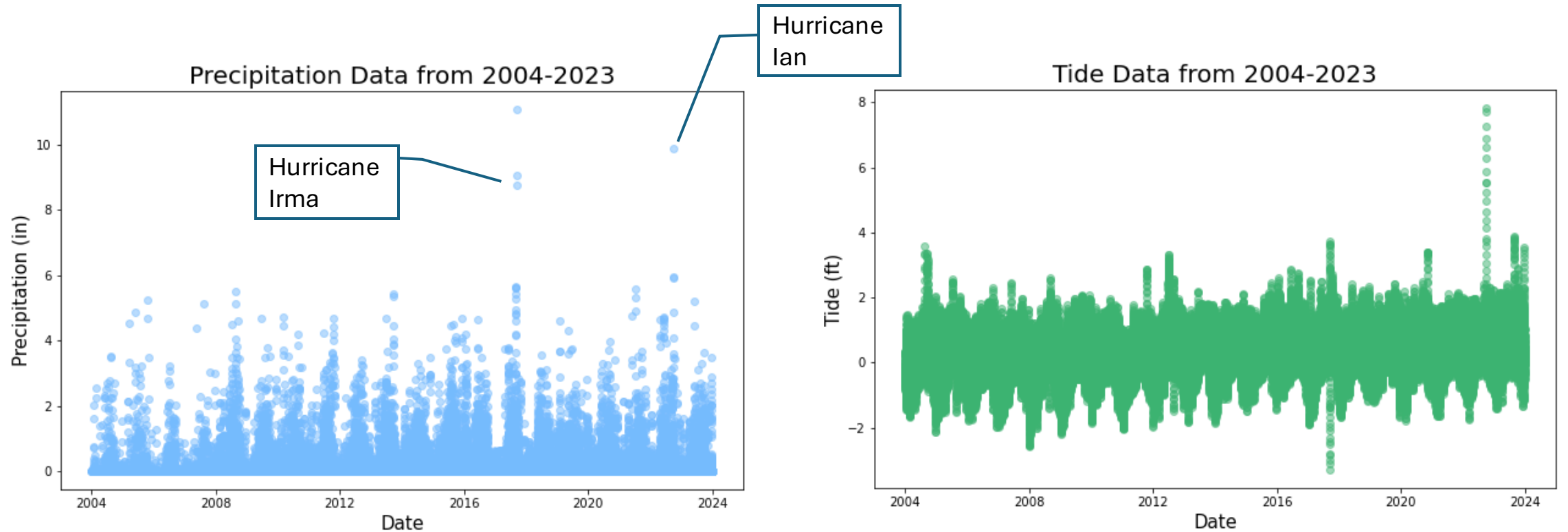
Average elevation: 4.6 ft above MSL

Minimum Elevation: 0.3 ft

Maximum Elevation: 17.1 ft



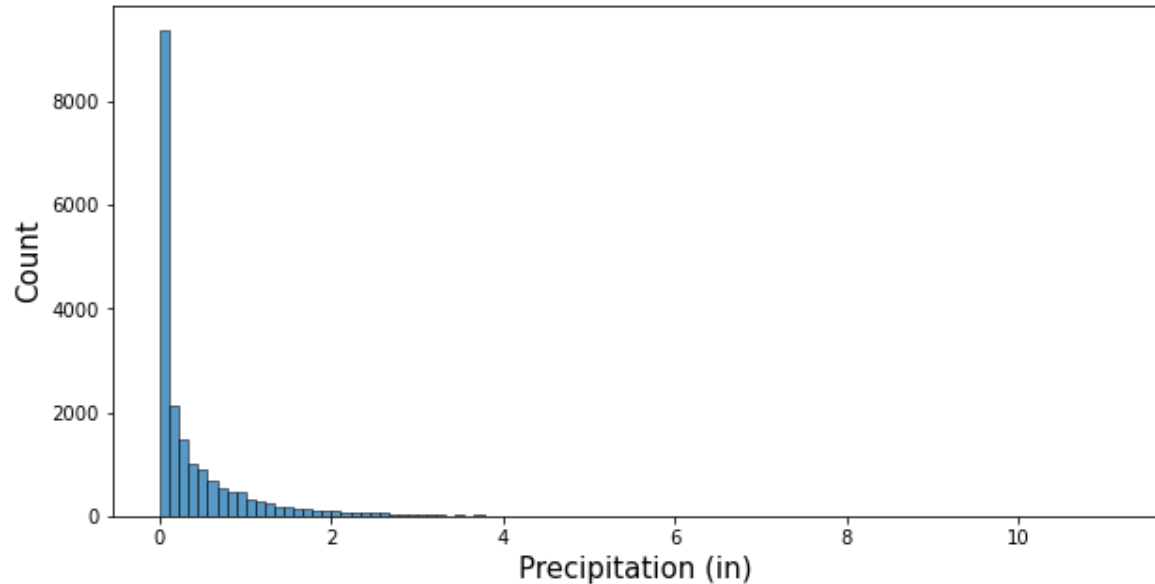
Original Precipitation and Tide Data Plots



The precipitation and tide datasets used in the project will only be from 2004 – 2023 to capture the last 20 years of data.

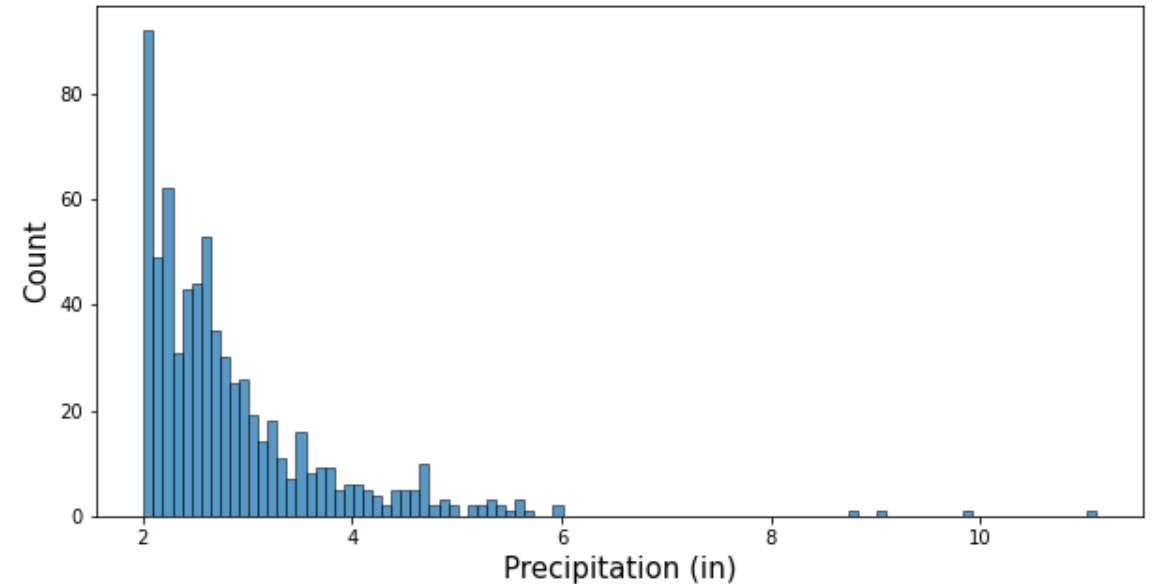
Most daily precipitation data is between 0 – 3 in.

Distribution of Daily Precipitation at All Stations



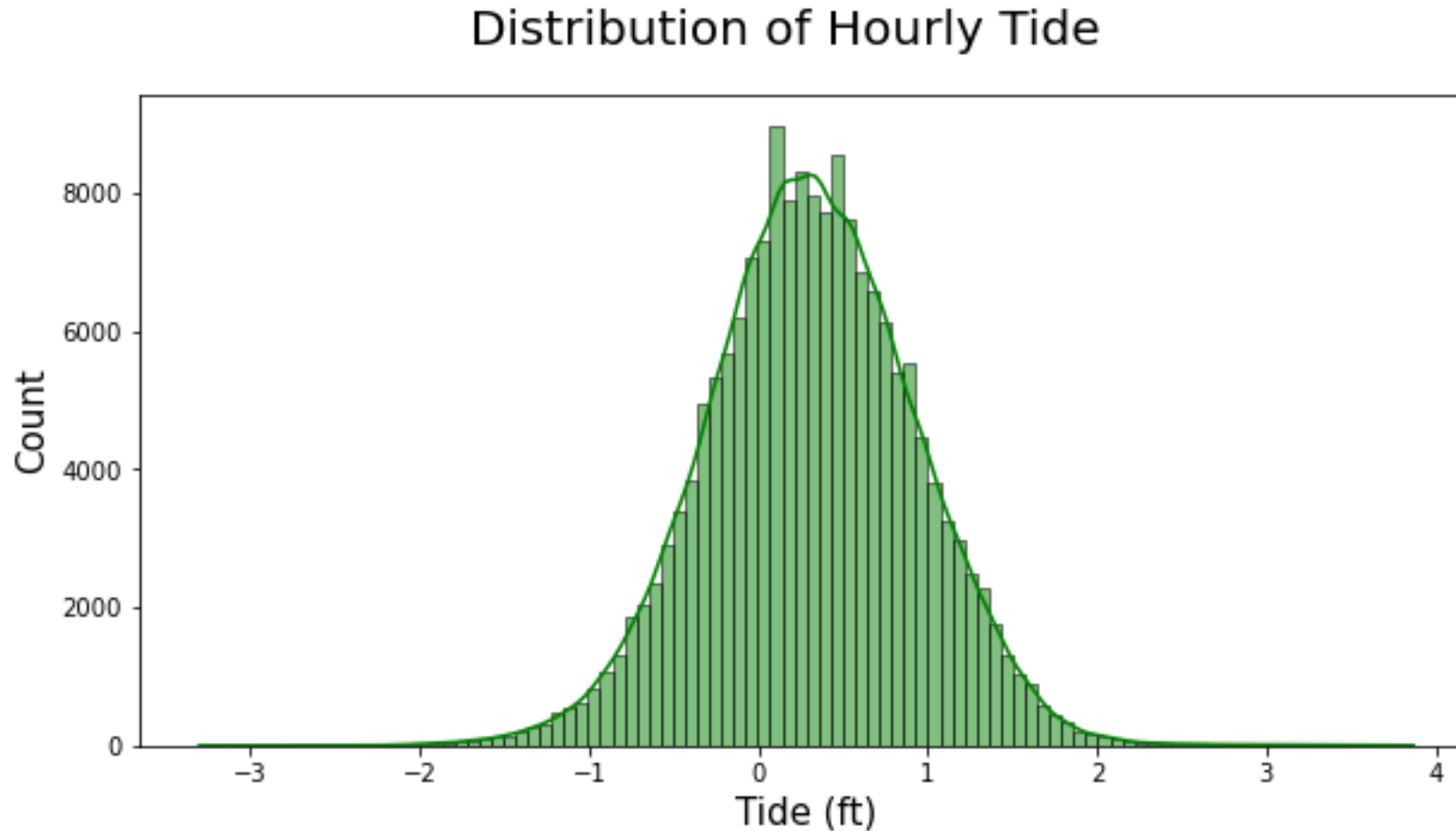
Only daily precipitation > 0 in. included in this histogram

Distribution of Daily Precipitation > 2 in. at All Stations



For this project, we will assume that “severe” precipitation is > 3 in. per day

Most hourly tide data is between -2 to +2 ft.
below/above mean sea level



Methodology to Data Quality Checking Process

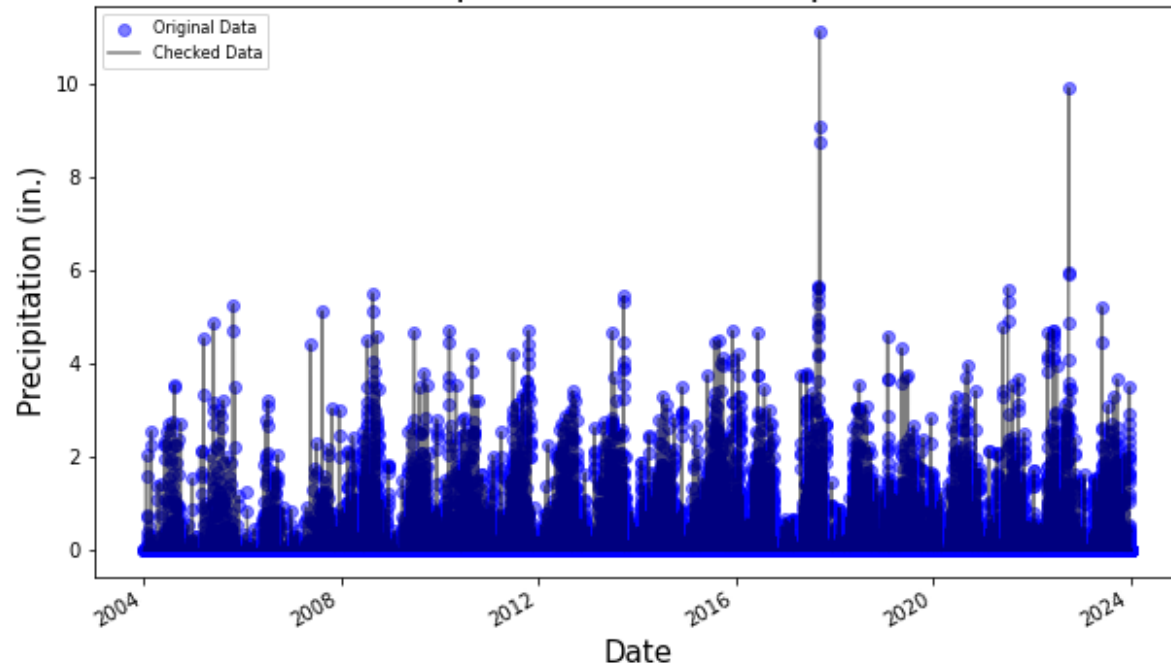
- Replace 'no data values' in precipitation data with Not a number (NaN)
 - From the Precipitation Data Metadata, the values 9999 means the data value is missing.
 - Replace all values 9999 with NaN.
- Replace trace precipitation values
 - Replace precipitation values with T (trace) flag that with 0.005 inches
- Check Blanks
 - Counts the number of Precipitation or Tide entries that are NaN or empty.
- Check Range
 - Check how many values are outside of the expected range.
 - 0 in. < Precipitation 25 in.
 - -4 ft. < Tide < 4 ft.

Data Quality Checking Results

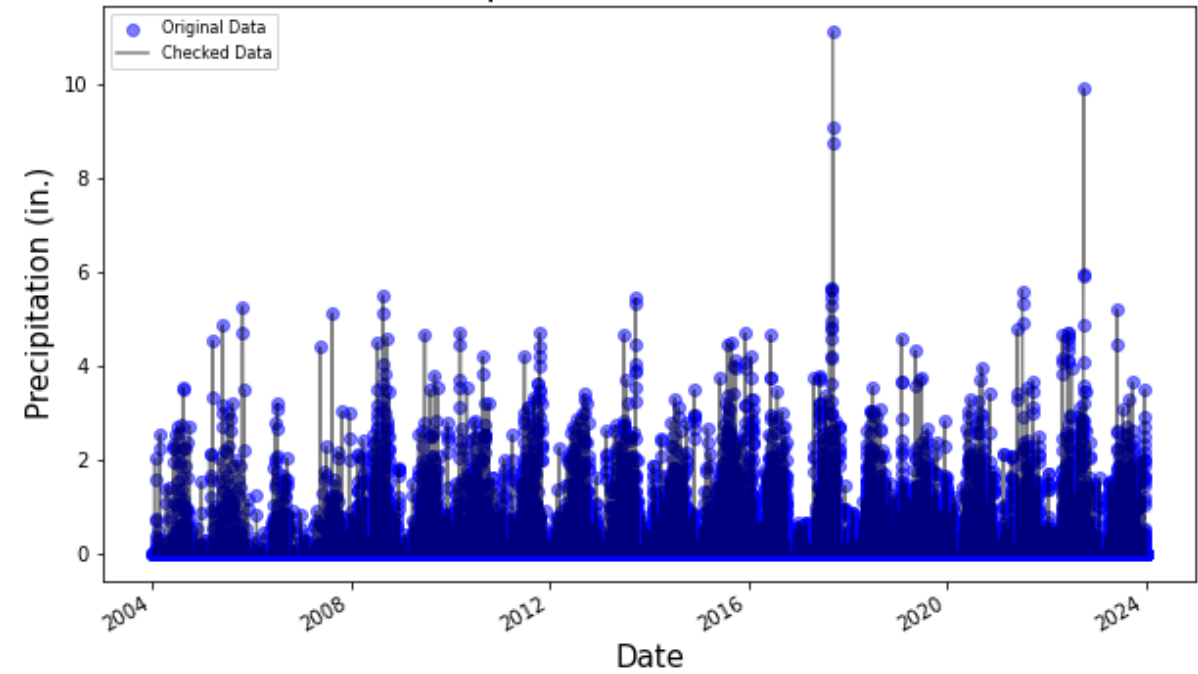
Check	Precipitation	Tide
1. Trace	2792	Not Applicable
2. No Data (9999)	0	Not Applicable
3. Blanks	609	0
4. Check Range	0	14

Data Quality Plots

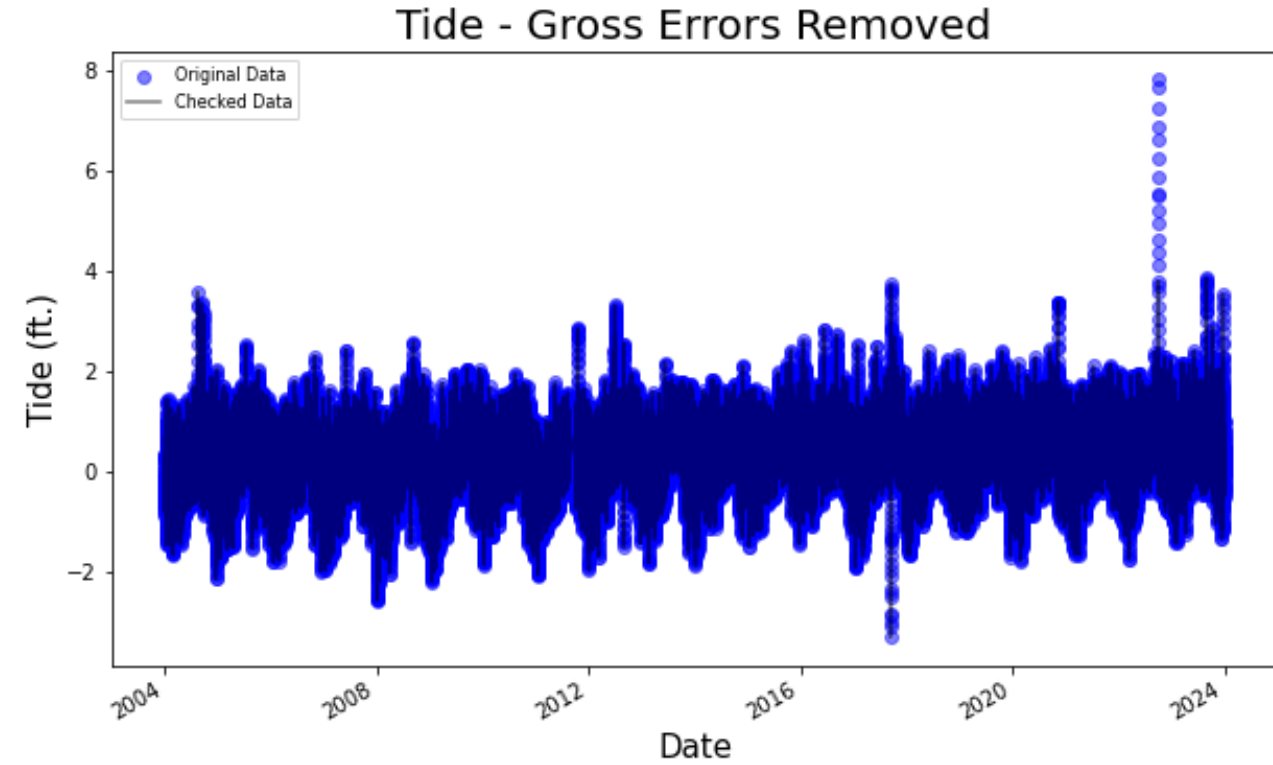
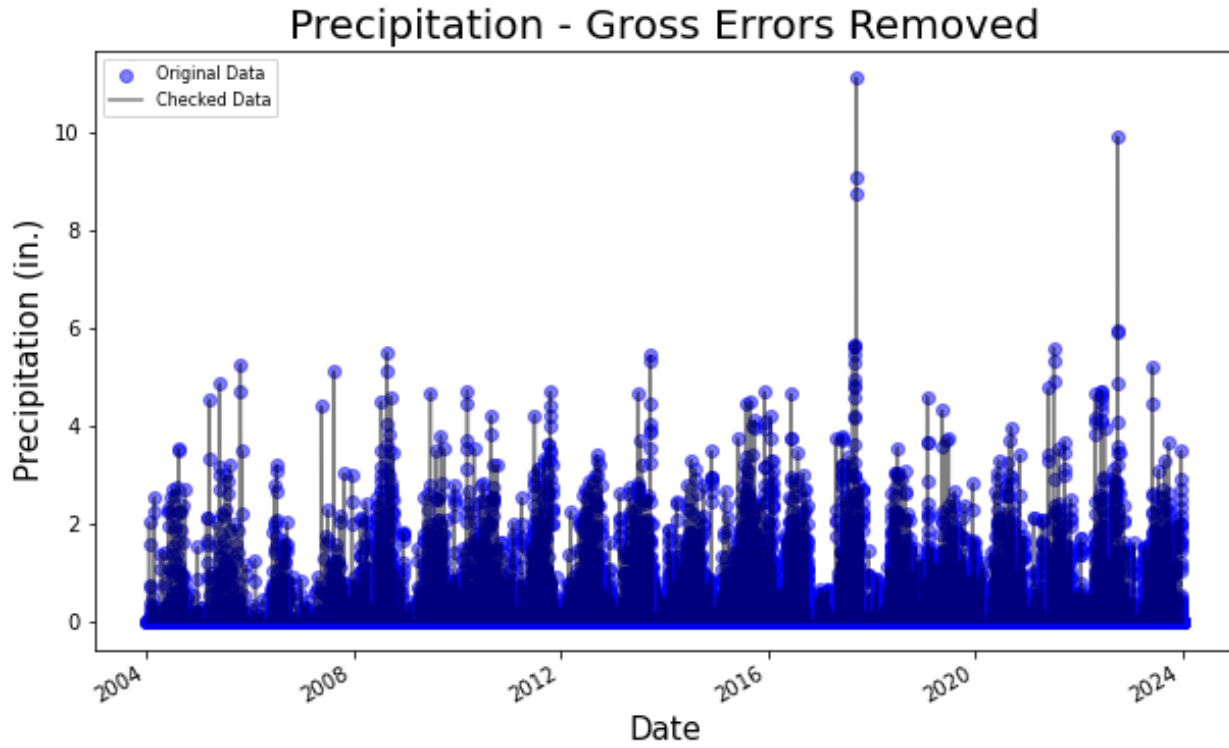
Precipitation - Trace Replace



Precipitation - 9999 Check

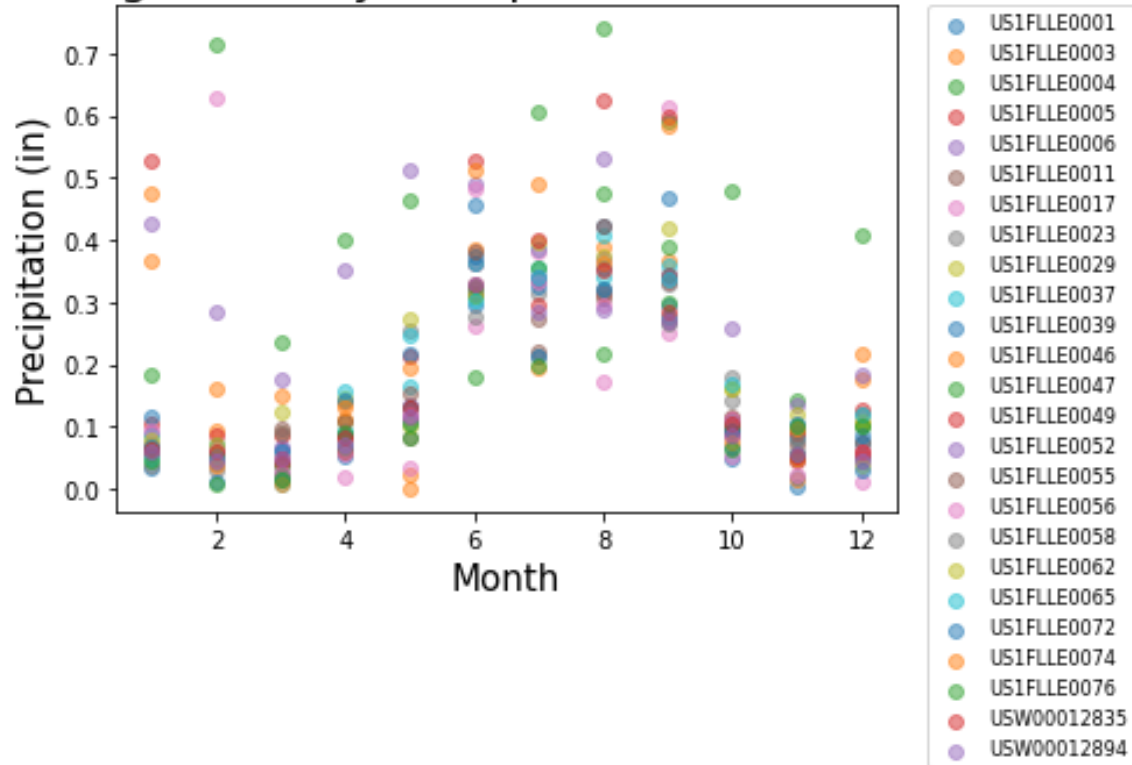


Gross Errors Removed

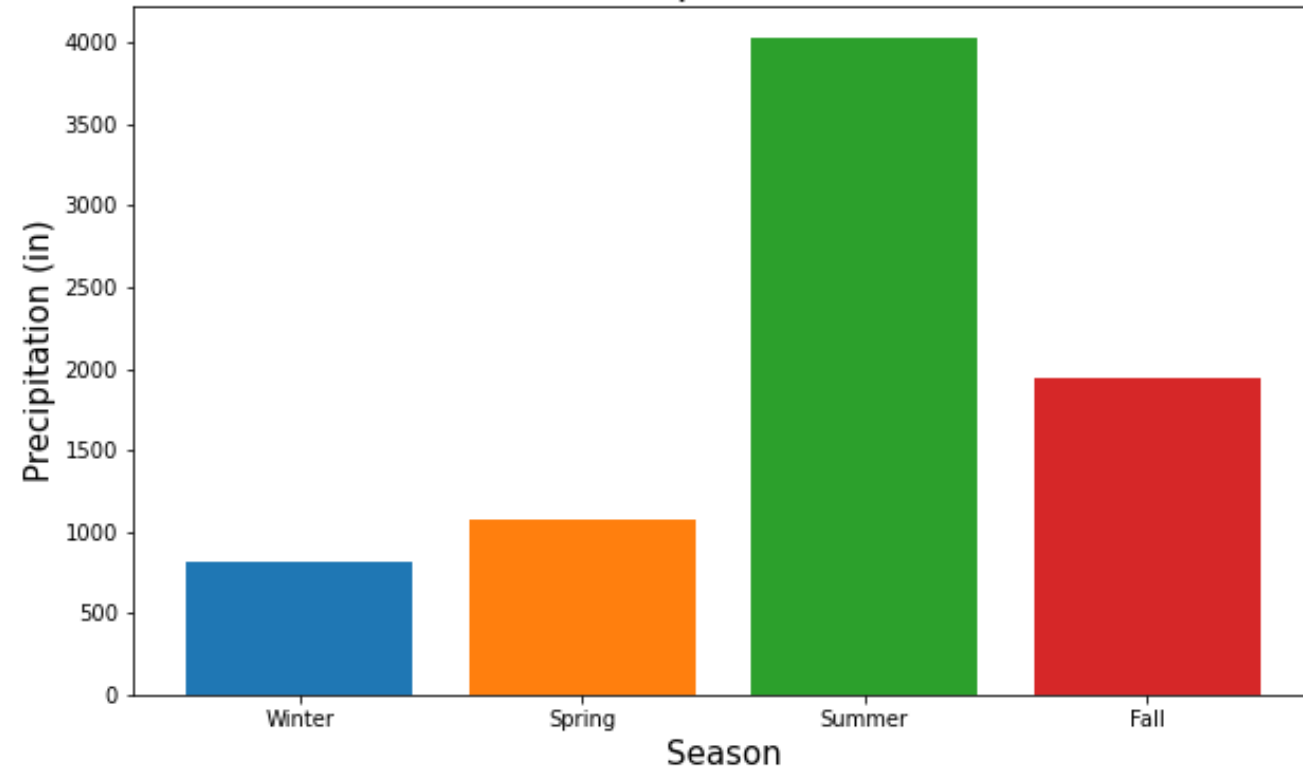


Statistical and Graphical Analysis Results

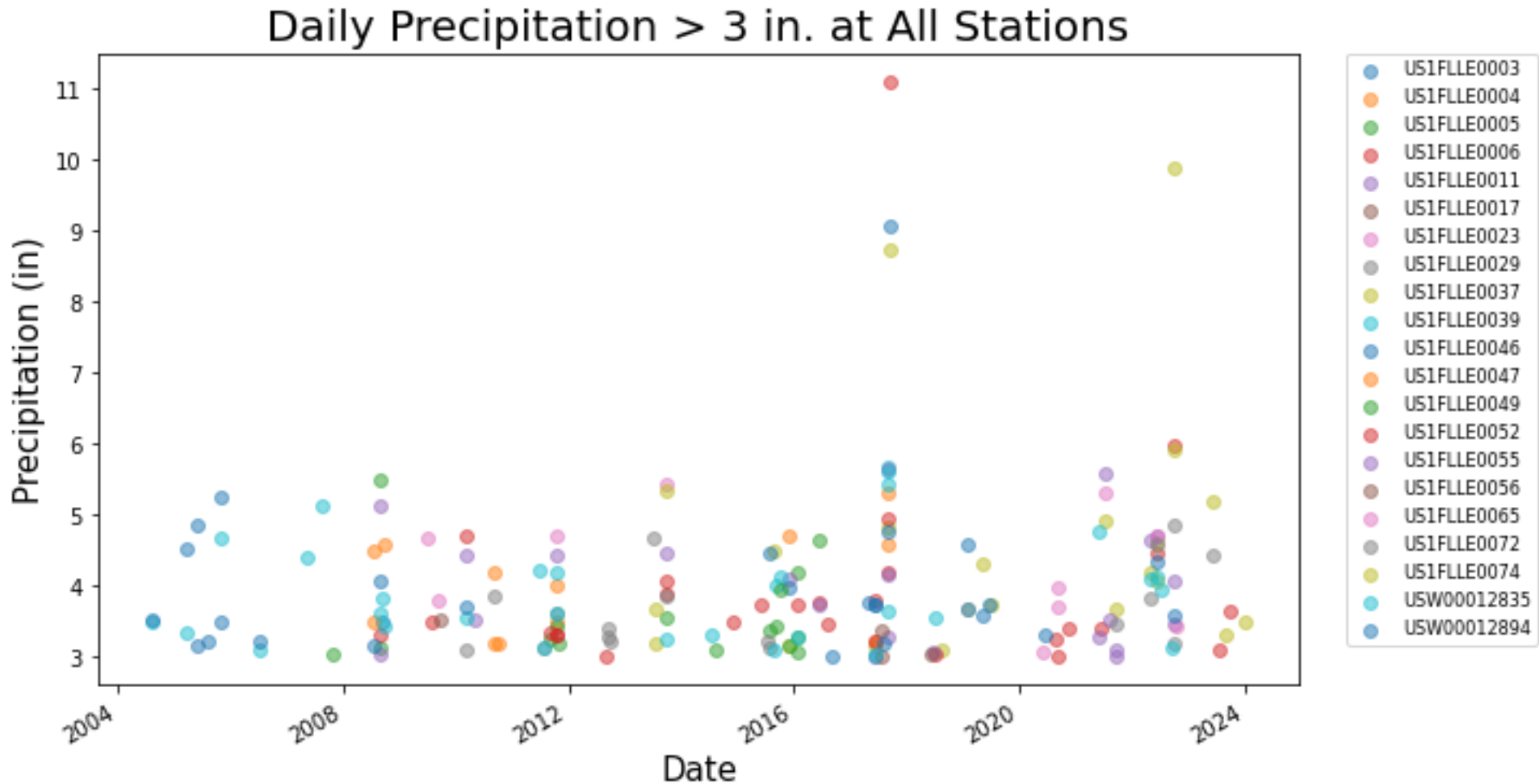
Average Monthly Precipitation at All Stations



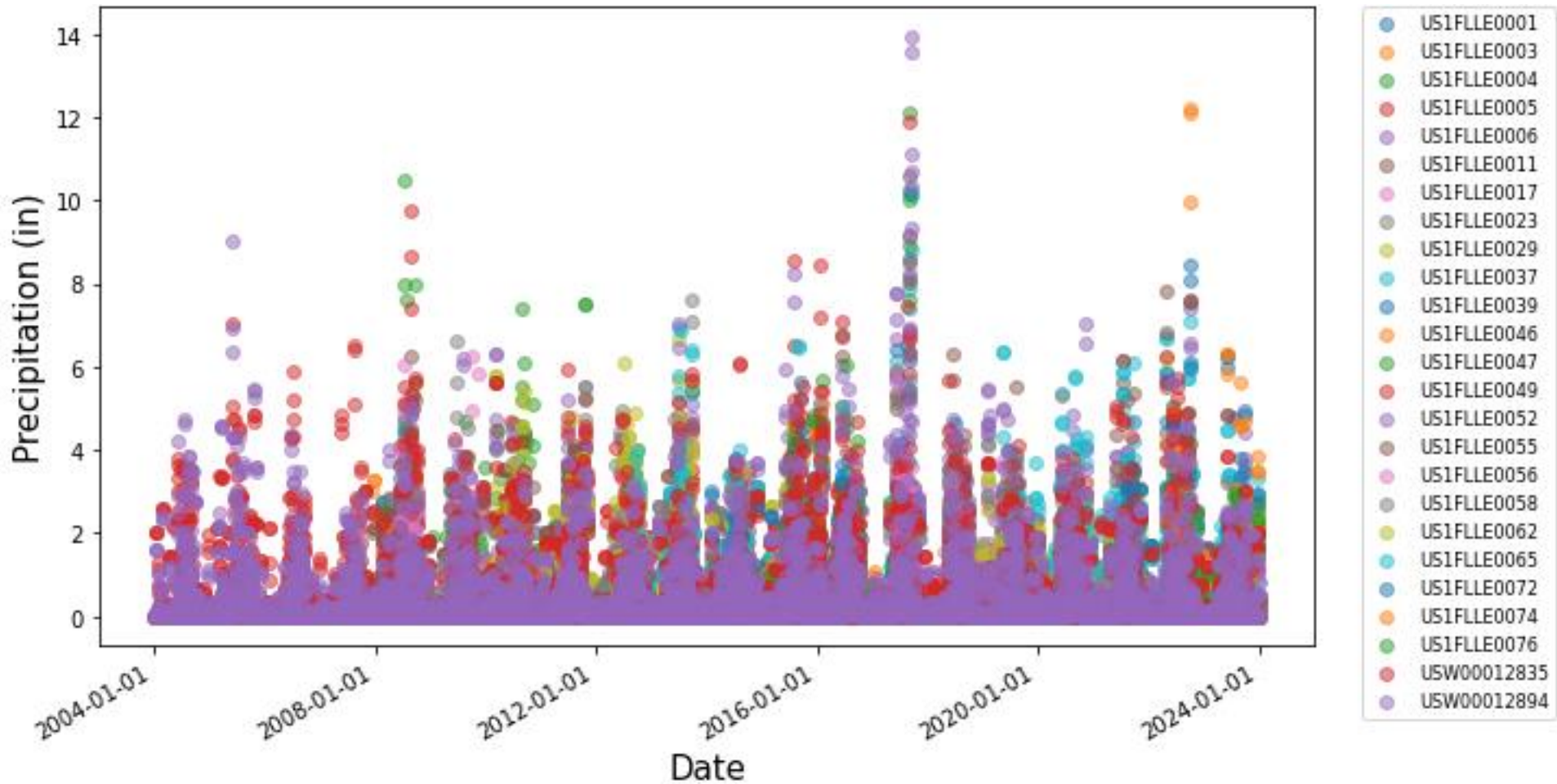
Sum of Seasonal Precipitation Across All Stations



Extreme Daily Precipitation

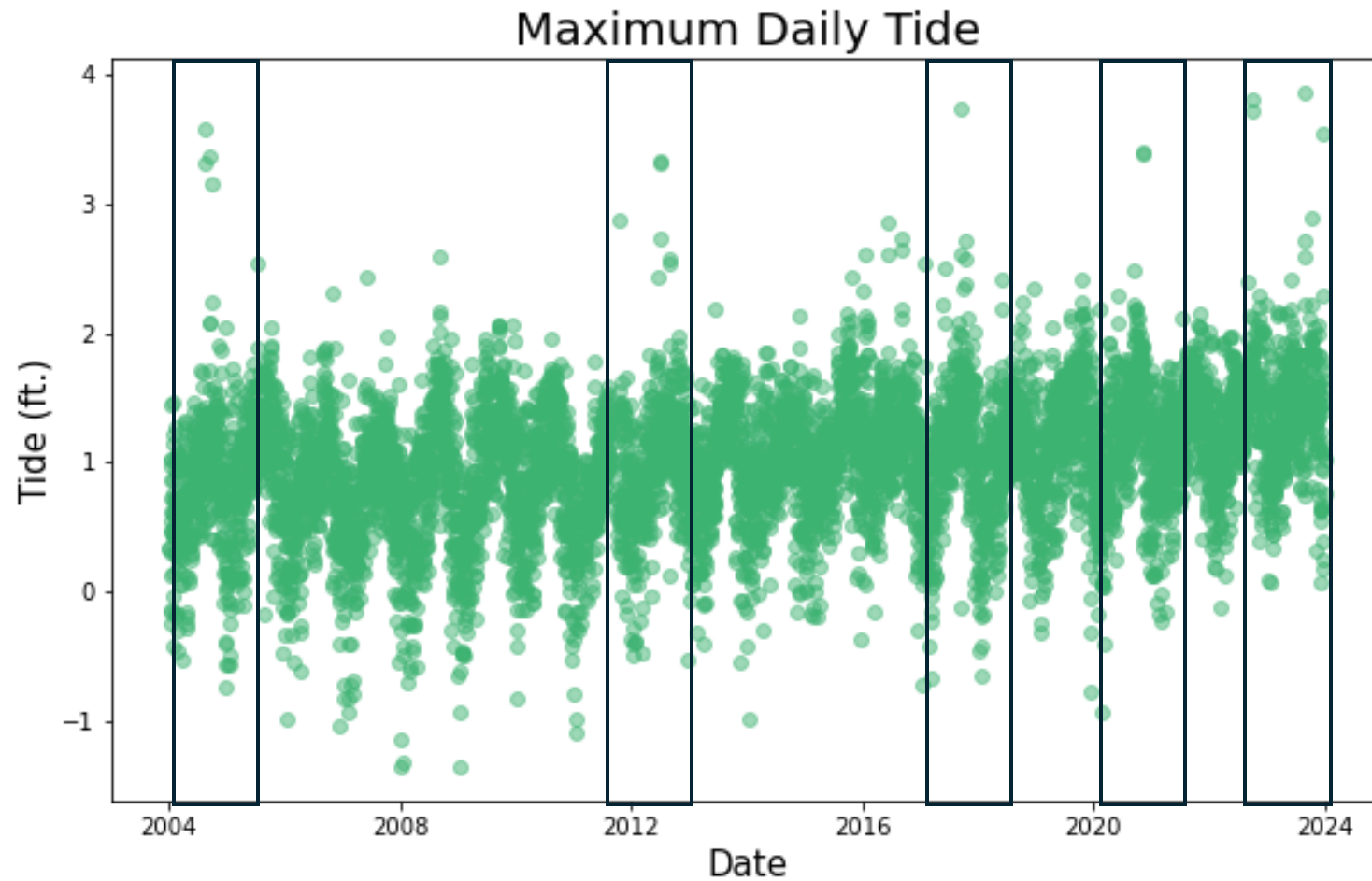


3-Day Rolling Sum of Precipitation at All Stations

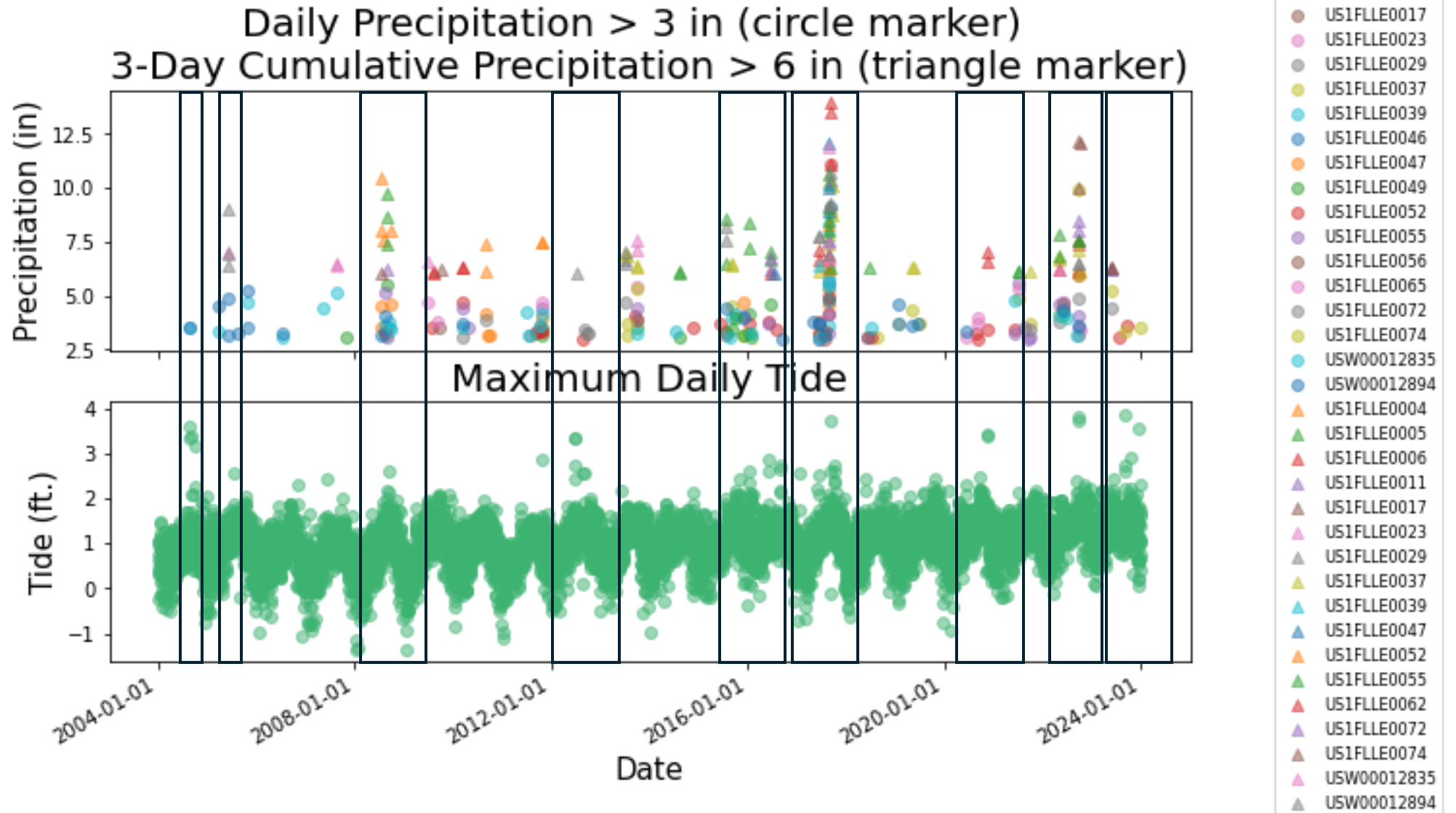


3-day cumulative precipitation is just as significant as extreme daily precipitation

High tide flooding regionally occurs more often during certain years



High Tide & Extreme Precipitation

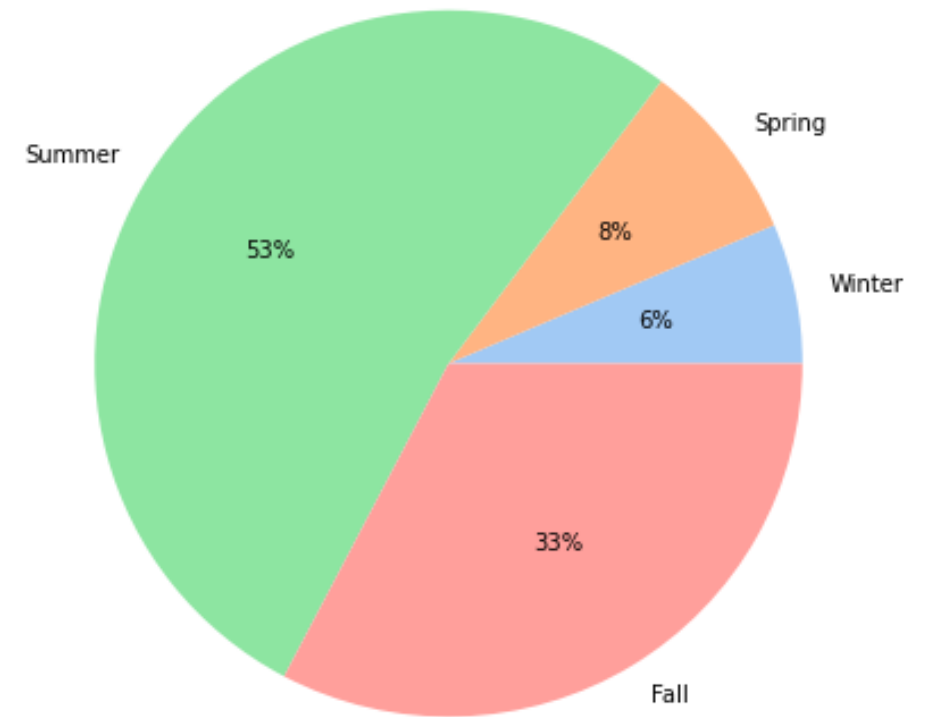
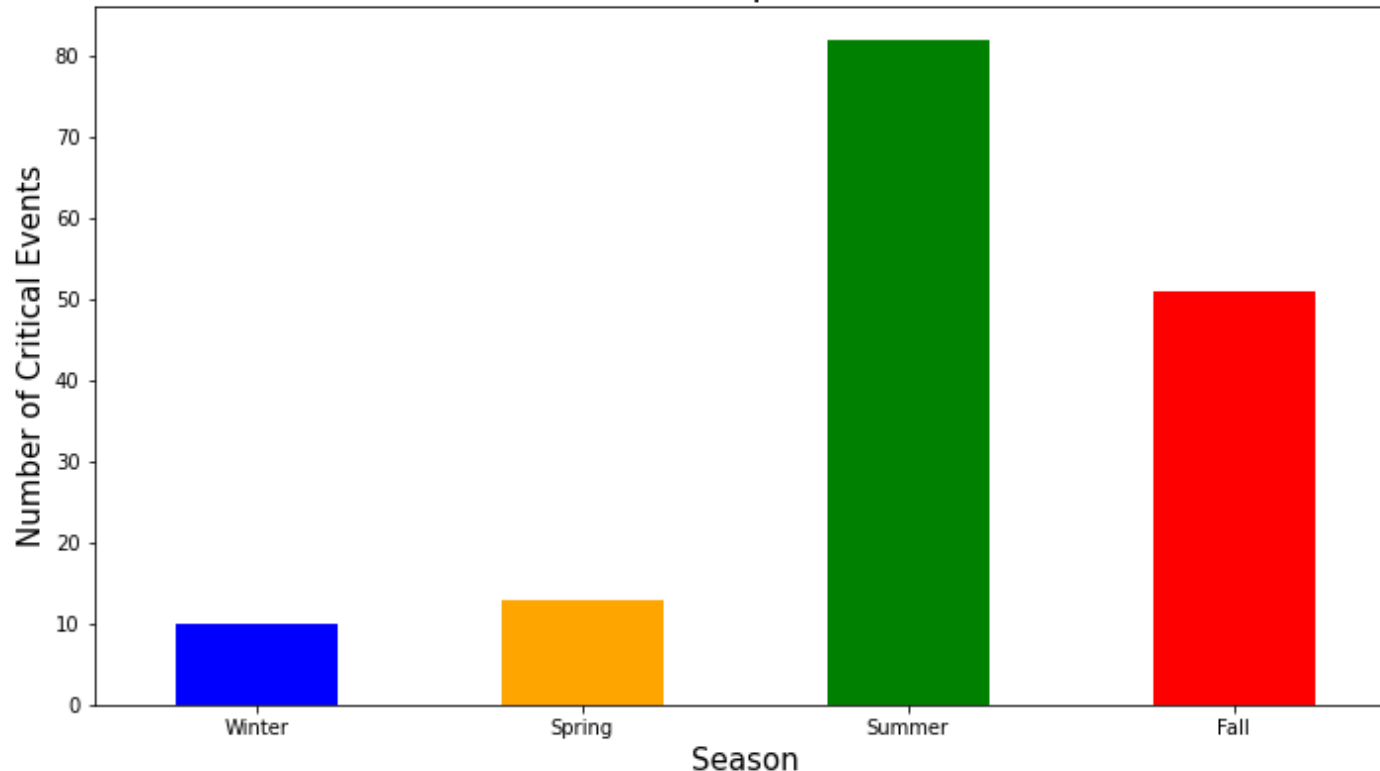


High tide and intense precipitation at Fort Myers occurs most often in the summer

Set parameters:

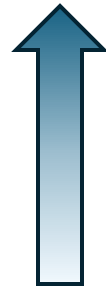
- Maximum Daily Precipitation > 3 in. and when Maximum Tide > 3 ft.
- 3-Day Rolling Precipitation > 6 in. and when Maximum Tide > 3 ft.

Number of Extreme Tide and Precipitation Coincidences in a Season



Summary of Data and Graphical Analysis

- The chances (from highest to lowest) for high precipitation and tide events to occur in Fort Myers, FL is:
 - Summer (June to August)
 - Fall (September to November)
 - Spring (March to May)
 - Winter (December to February)



High chance of critical events

Conclusion

- What we learned that would benefit future research
 - Upscaling this method will require more coding
 - This method may encounter issues for locations that have snow
 - We need to find flooding data for our hypothesis to be fully answered
- Future applications
 - Flooding due to rising sea levels and higher frequency of rain events in the southeast U.S. may be of a greater concern under a warming climate.
 - This method can only be applied to coastal cities that have available NOAA precipitation and tide datasets.



Thanks for listening!

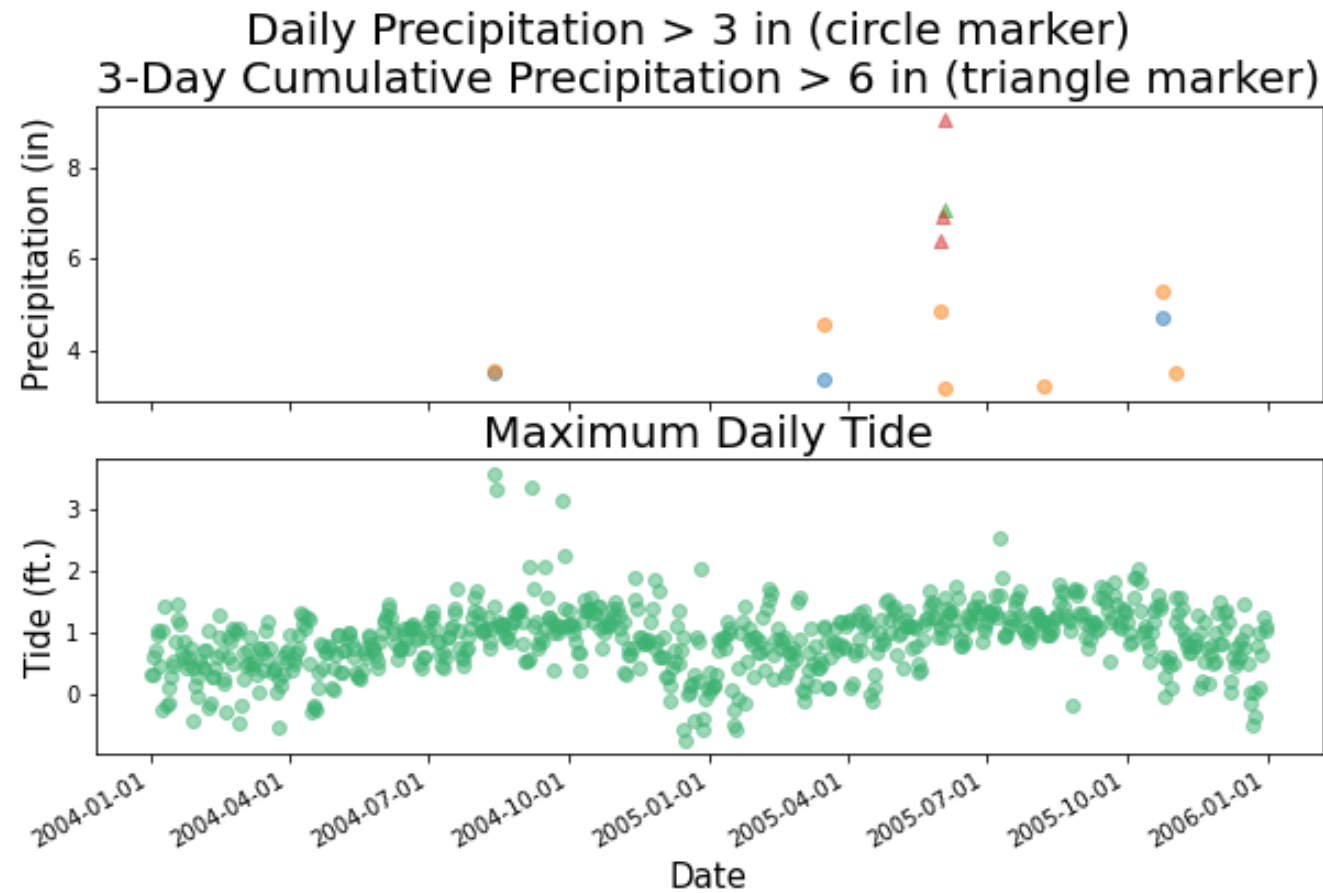
- Shoutout to our advisor, Dr. Cherkauer, for his help and guidance on our project 😊
- Questions?
 - Reach out to us at mbinnase@purdue.edu and yamm@purdue.edu

References

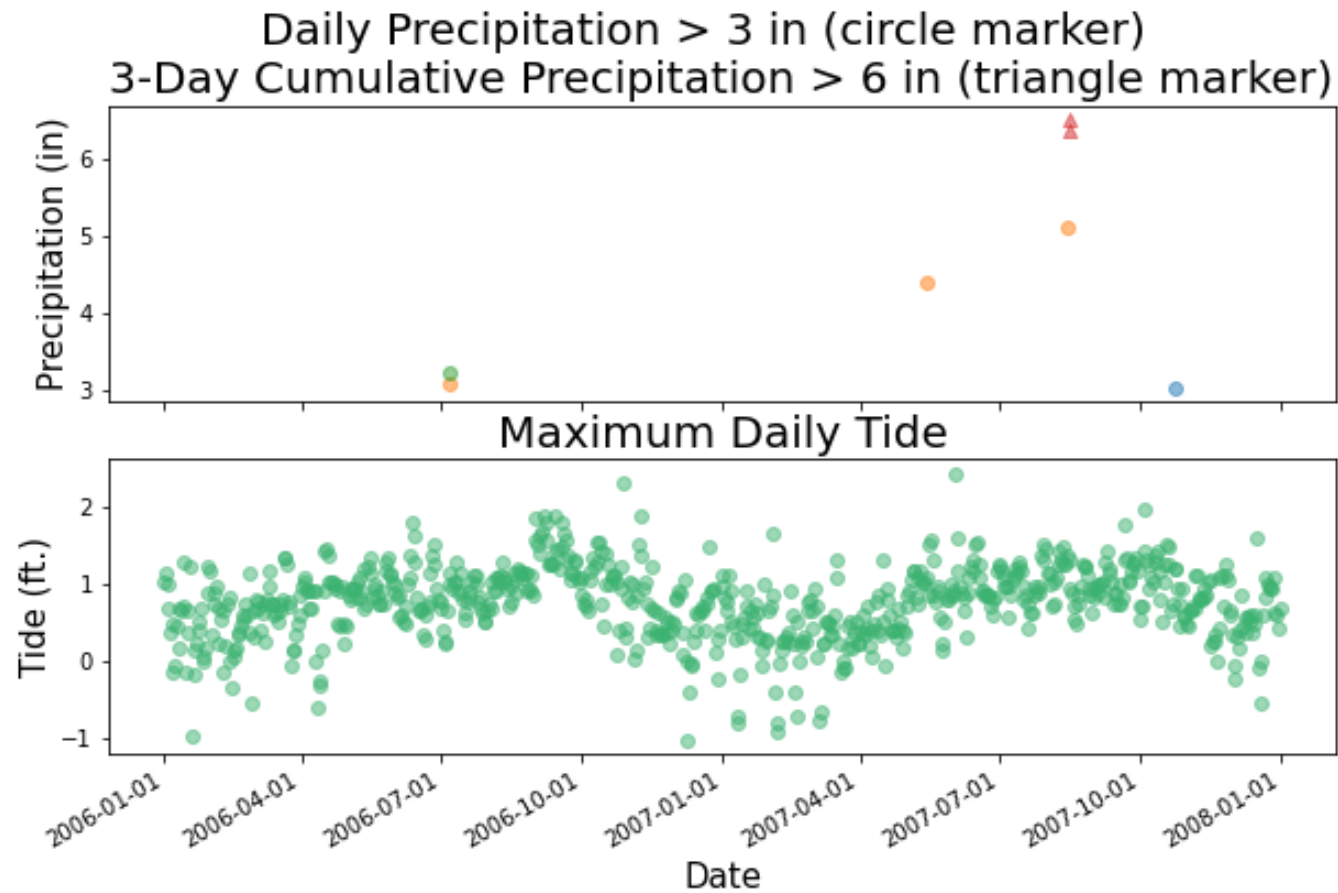
- Ebi, K.L., J.M. Balbus, G. Luber, A. Bole, A. Crimmins, G. Glass, S. Saha, M.M. Shimamoto, J. Trtanj & J.L. White-Newsome. (2018). *Human health*. In: *Impacts, risks, and adaptation in the United States: Fourth national climate assessment, volume II* [Reidmiller, D.R., C.W. Avery, D.R. Easterling, K.E. Kunkel, K.L.M. Lewis, T.K. Maycock, and B.C. Stewart (eds.)]. U.S. Global Change Research Program, Washington, DC, pp. 544, 551–552. [doi: 10.7930/NCA4.2018.CH14](https://doi.org/10.7930/NCA4.2018.CH14)
- Sweet, William (William VanderVeer) et al. (2018). *Patterns and projections of high tide flooding along the U.S. coastline using a common impact threshold*. <http://doi.org/10.7289/V5/TR-NOS-COOPS-086>

Extra Slides

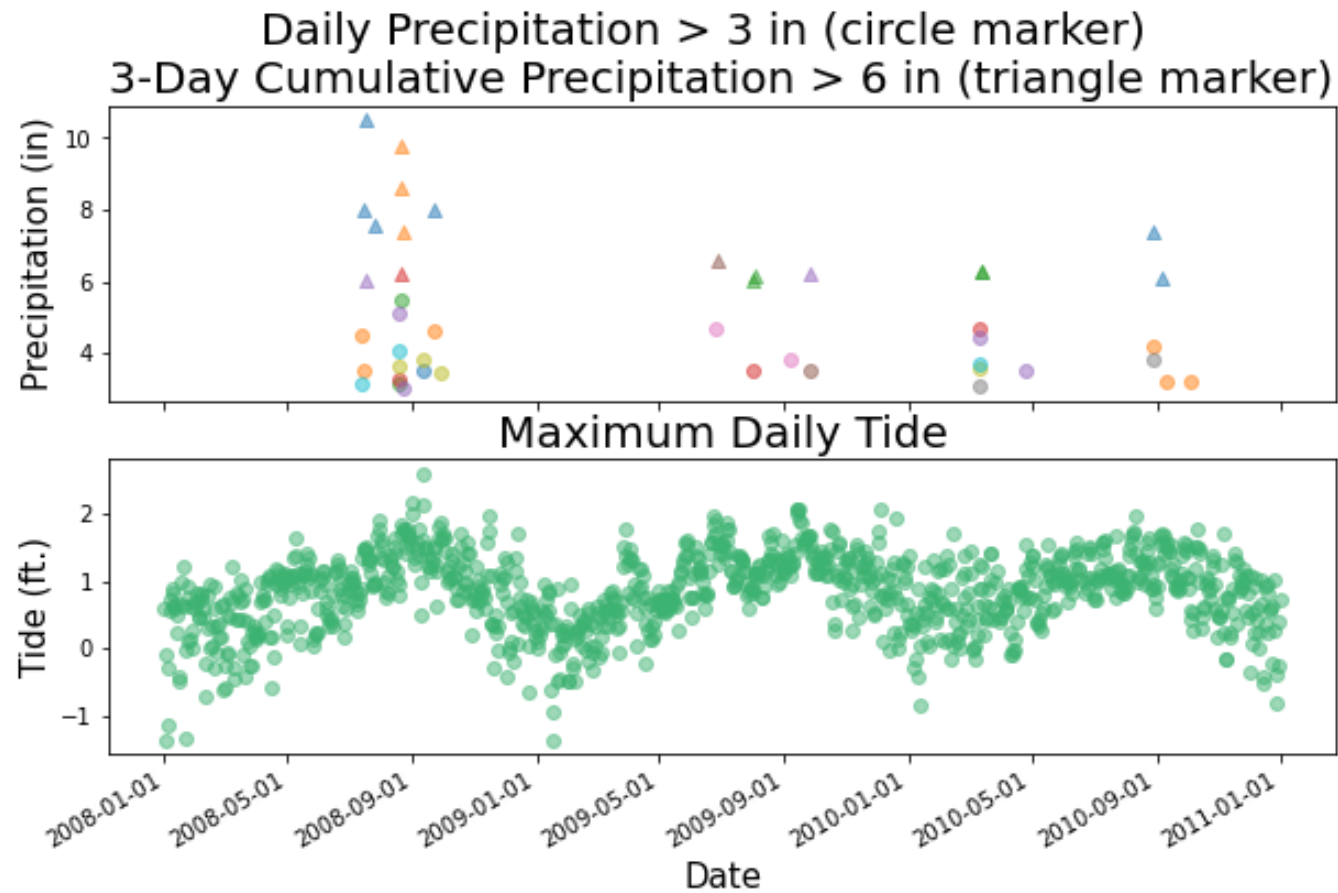
2004 - 2005



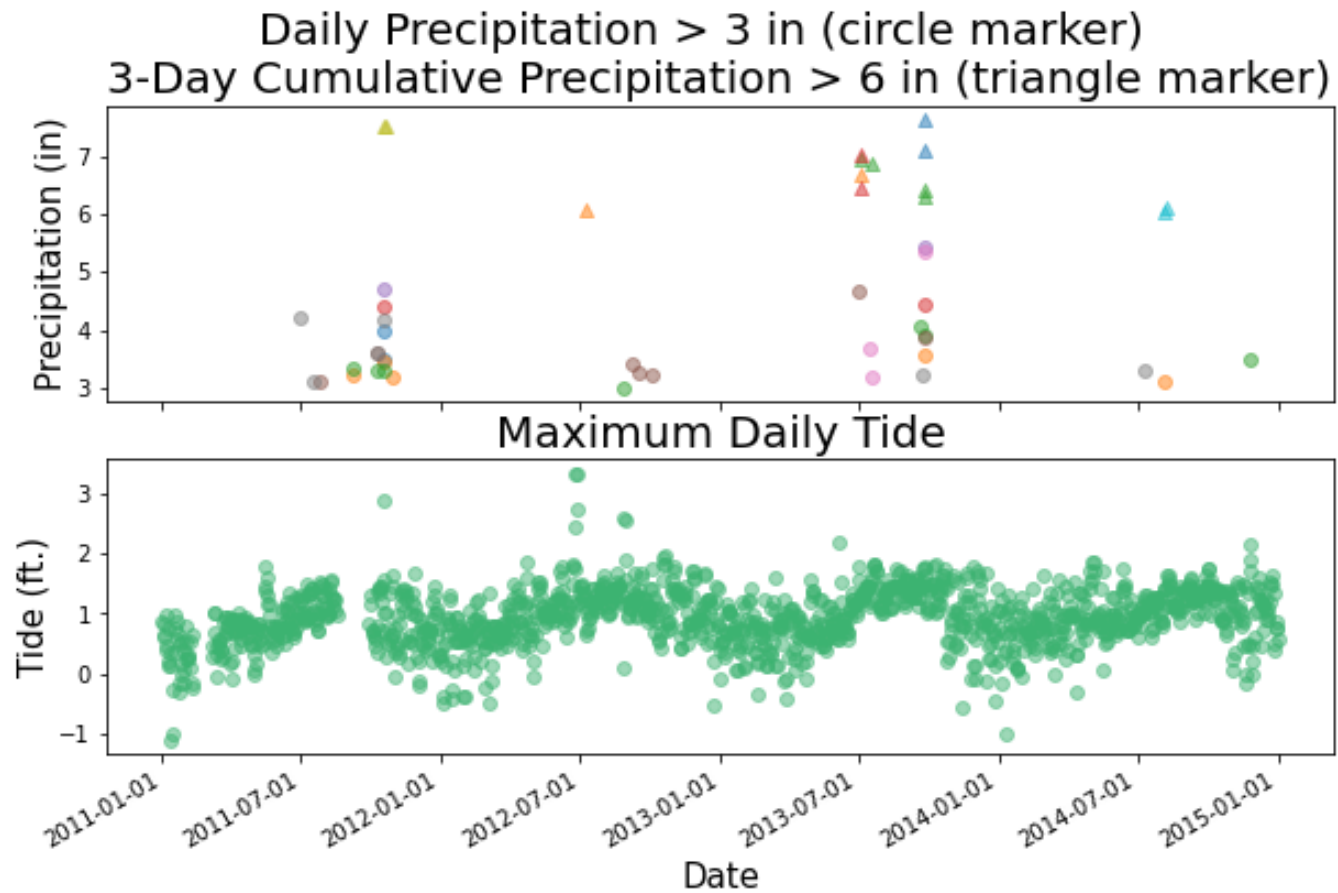
2006 - 2007



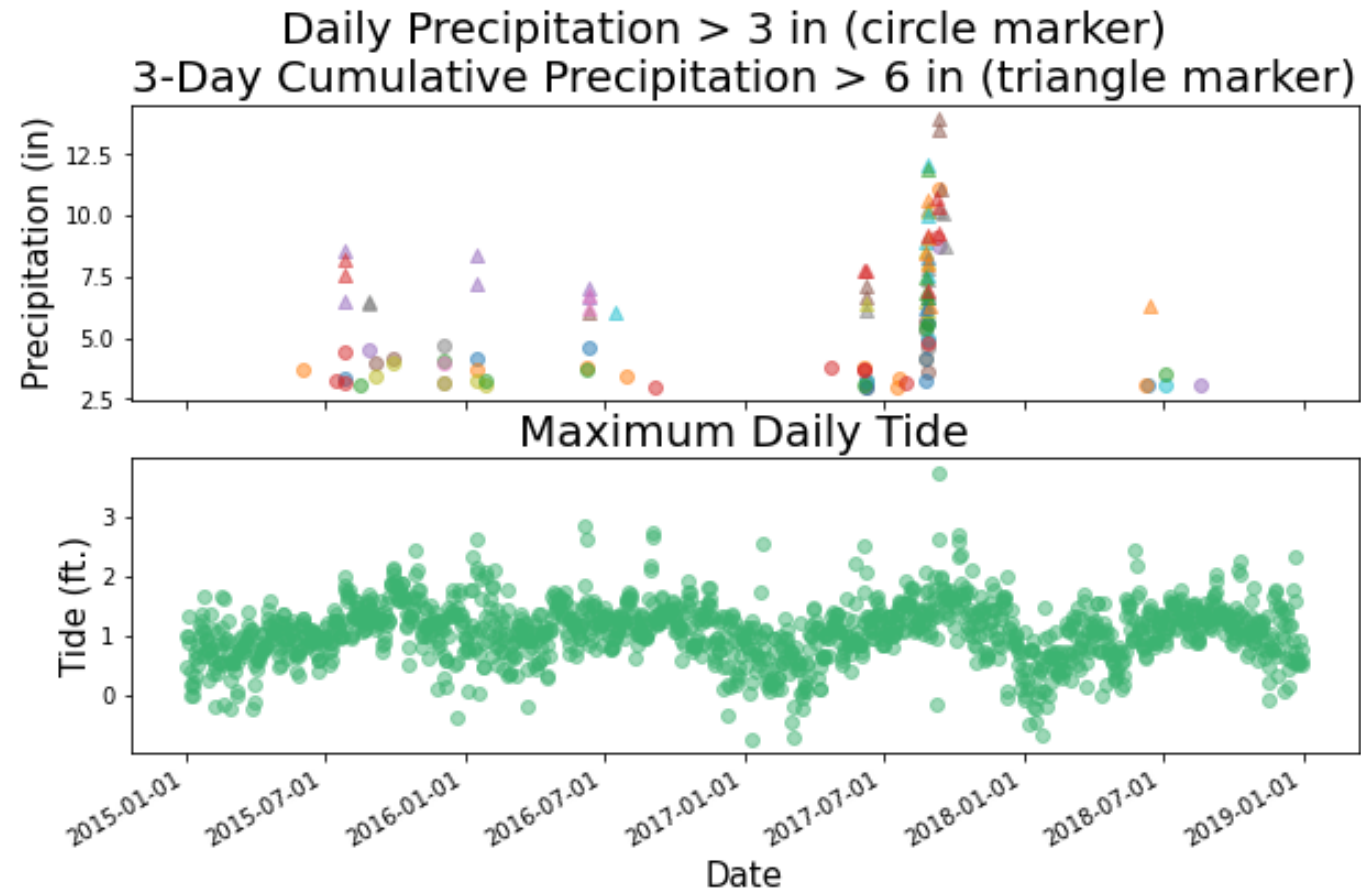
2008 - 2010



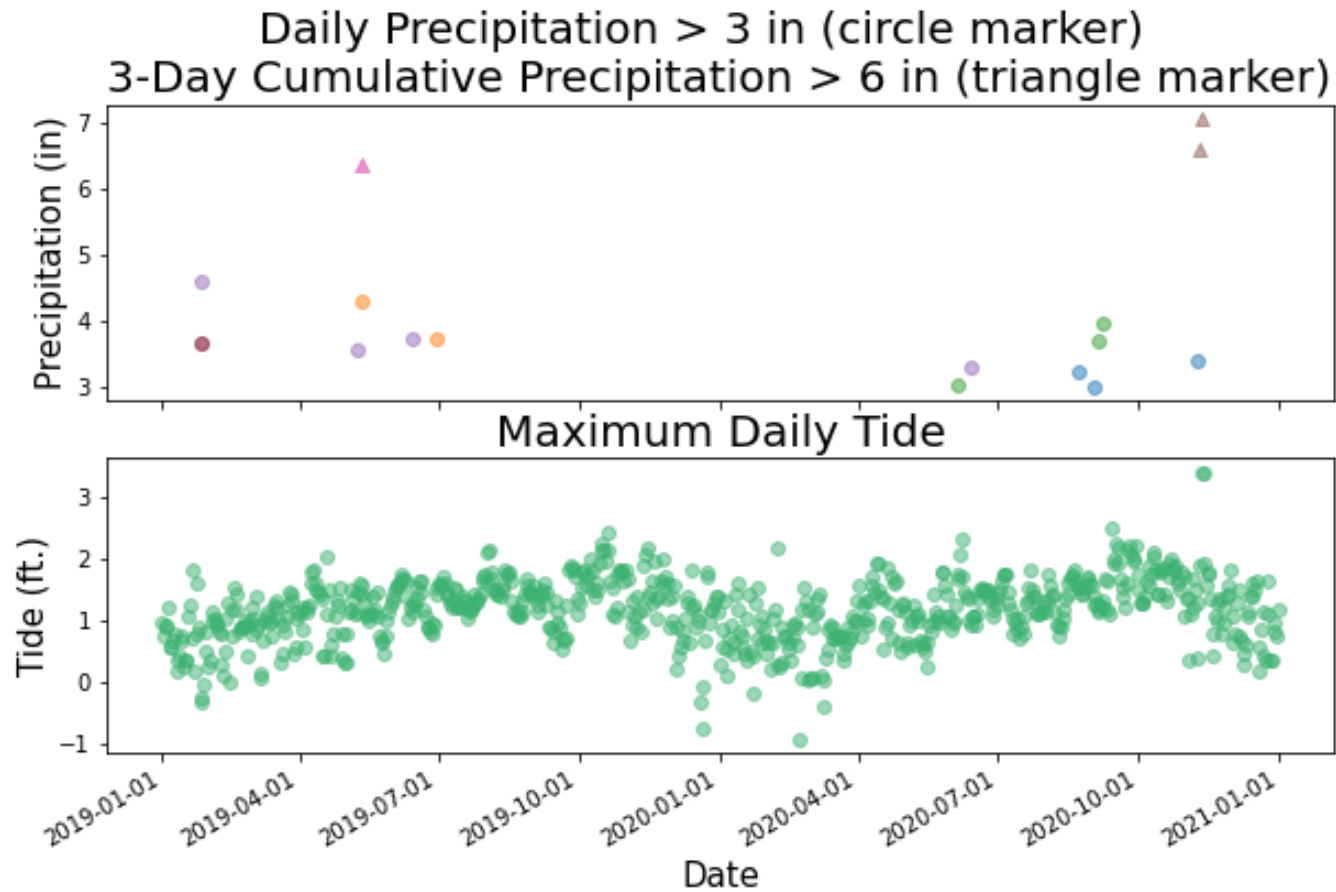
2011 - 2014



2015 - 2018



2019 - 2020



2021 - 2023

